

Shaping quantum collisions with light

Optimal laser control of molecule formation through an optical Feshbach resonance

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THE QUESTION

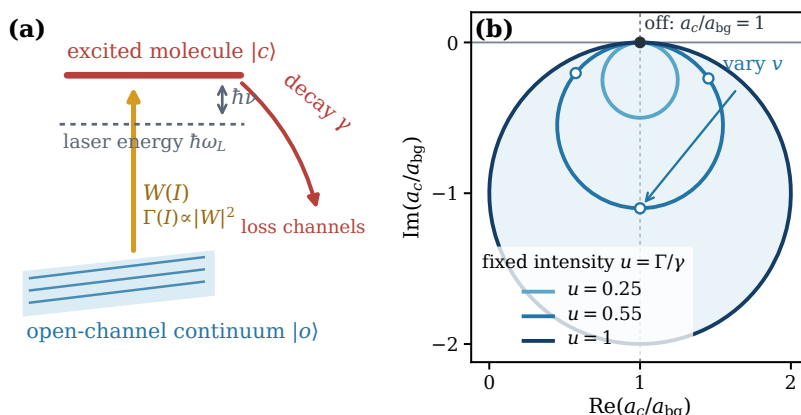
Can a laser assemble molecules from an ultracold gas without simply destroying the atoms?

The problem in plain language One laser, two inseparable effects

At a few billionths of a degree above absolute zero, atoms behave as coherent matter waves. A laser tuned near a molecular resonance can change how two waves scatter—and can briefly couple an atom pair to a molecular state.

That opportunity comes with a cost: the excited molecule can decay. The useful interaction and the unwanted loss are two parts of the same *complex scattering length*. Turning the intensity up does not provide an independent “molecule knob.”

The gas also remembers. The rate at one instant depends on pair correlations built by the entire earlier pulse, so the best control is a time sequence rather than one fixed laser setting.



Left: the laser couples colliding atoms to an excited molecule; spontaneous decay opens a loss channel. **Right:** at fixed intensity, changing detuning moves the scattering length around a circle in the complex plane. Intensity selects the circle. The elastic and lossy responses therefore cannot be programmed independently.

The central trade-off

Detuning from resonance reduces decay, but also weakens the interaction. Intensity strengthens both. The optimization must decide **when to build correlations** and **when to convert them into products**.

Why use ^{88}Sr ?

Its narrow 689 nm intercombination transition enables optical control with a comparatively long-lived excited molecular state. Its background interaction is close to zero, so changes of only a few Bohr radii strongly affect the gas [8, 9].

From two laser waveforms to one objective

1

Laser controls

$\Gamma(t)$ sets coupling;
 $\nu(t)$ sets detuning.

2

Complex interaction

$a_c(t)$ links elastic
response and loss.

3

Quantum memory

A Volterra equation
evolves $\eta(t)$.

4

Correlated pairs

Tan's contact
scales as $|\eta(t)|^2$.

5

Molecular yield

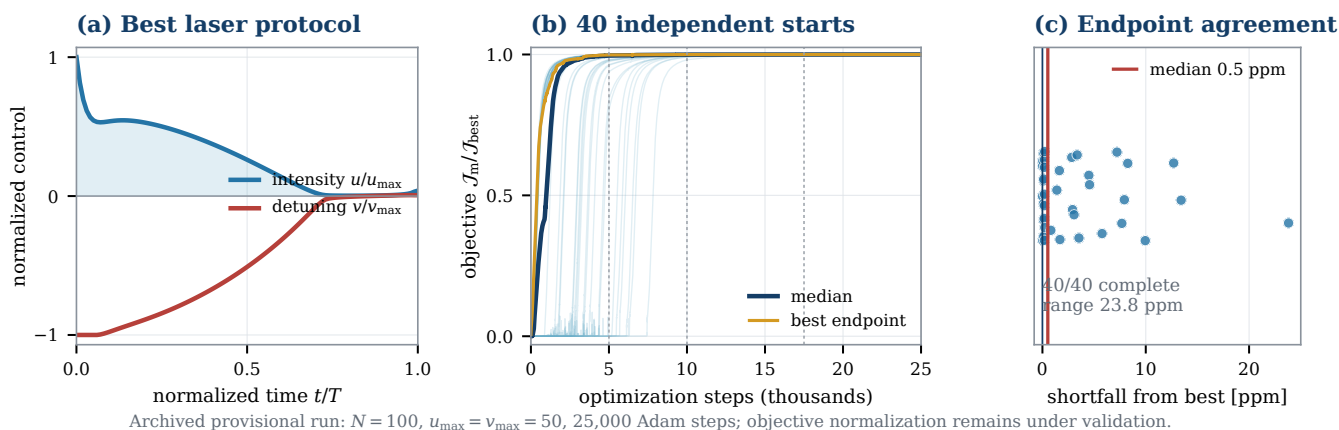
Accumulate the
lossy pair flux over
time.

Strategy: use automatic differentiation to search smooth, bounded intensity and detuning histories—while retaining the memory of every earlier control decision.

What the calculation shows so far

A completed archived multi-start run provides a concrete pulse and a reproducibility check.

PRELIMINARY
VALIDATION IN PROGRESS



40/40

archived starts completed

25,000

Adam steps per start

23.8 ppm

best-to-worst objective
spread

$N = 100$

provisional time grid

How to read the pulse

- 1. Couple strongly at first.** The best archived intensity begins at the cap, then quickly drops.
- 2. Chirp the detuning.** The pulse begins on one far-detuned edge and moves smoothly toward resonance.
- 3. Finish gently.** Both controls approach zero near the end. This is consistent with a prepare-convert sequence, but that mechanism still needs to be confirmed from $|\eta(t)|^2$ and the instantaneous flux.

Experimental scale

689 nm narrow 1S_0 - 3P_1 intercombination line.

$2\pi \times 15$ kHz excited-molecule natural linewidth.

-24 MHz binding of the demonstrated photoassociation level.

0.057 W cm^{-2} published comparison intensity, about 0.47 mW for a $725 \mu\text{m}$ Gaussian waist.

Experiments changed the ^{88}Sr scattering length by up to approximately $\pm 10 a_{\text{bg}}$ with millisecond sample lifetimes [9].

Before claiming an optimum

Still required:

- reconcile the exact noninteracting initialization and kernel coefficient;
- show time-grid convergence beyond $N = 100$;
- derive the prefactor connecting the objective to a molecule fraction;
- complete the laser-cap sweep and identify any saturation regime;
- test sensitivity to seeds, pulse smoothing, loss calibration, and detuning error.

The graph above therefore demonstrates *optimizer agreement*, not yet experimental yield accuracy.

Take-home message: optical molecule formation is a control problem with memory. The useful interaction and the loss channel are inseparable, so timing and detuning can matter as much as peak laser power.

PART II / TECHNICAL REPORT AND DERIVATION

Optimal Laser-Constrained Control of Molecular Formation from a Quantum Gas via Optical Feshbach Resonances

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Abstract

Optical Feshbach resonances provide a fast, species-flexible means of controlling ultracold interactions, but the same optical coupling that changes the real scattering length also opens an inelastic molecular channel. This report develops a numerical optimal-control framework for maximizing the molecular yield accumulated over a fixed time while respecting a finite laser-intensity budget. The physical controls are the dimensionless stimulated linewidth and detuning, which jointly determine a complex scattering length. Their history drives a universal short-time pair amplitude through a half-order Volterra equation, and the loss-contact relation converts that amplitude into a molecular production rate. The resulting constrained objective is optimized with automatically differentiated, GRAPE-style piecewise controls and translated to the 689-nm 1S_0 - 3P_1 optical Feshbach resonance of bosonic ^{88}Sr . The archived $N = 100$ multi-start result shown here is provisional; the absolute yield normalization, time-grid convergence, cap sweep, and laboratory prescription remain to be validated.

Keywords: optical Feshbach resonance, ultracold quantum gas, Tan contact, molecular formation, optimal control, Volterra equation, JAX

Technical-section status. The following pages preserve the detailed physics, derivations, numerical-method specification, results structure, and experimental translation. Items marked **[TO COMPLETE]** are deliberate verification tasks for the final report; numerical placeholders should be replaced only from archived, source-labelled artifacts.

1. INTRODUCTION

Feshbach resonances make the interaction strength of an ultracold gas a controllable experimental parameter. At low collision energy, this control is summarized by the s -wave scattering length, and resonant coupling can change both its magnitude and sign [7]. Optical coupling adds speed and spatial selectivity, including for species without useful magnetic resonances, at the cost of spontaneous-emission and photoassociation loss [1, 5].

The central difficulty is dynamical. Molecular production at one instant depends not only on the laser settings at that instant, but also on the short-range pair correlations accumulated over the entire preceding protocol. Tan's contact organizes these correlations [4, 6], while the universal short-time theory of Qi, Shi, and Zhai reduces their evolution to a history-dependent equation for a pair amplitude $\eta(t)$ [10]. This combination turns molecular formation into an optimal-control problem with memory.

Central research question

For a fixed observation time T and a maximum available laser coupling $\Gamma_{\max}$, which experimentally realizable intensity and detuning trajectories maximize the accumulated molecular yield, and when does increasing $\Gamma_{\max}$ cease to improve that yield?

1.1. Motivation and scope

Abstract control of the real and imaginary parts of a complex scattering length can generate protocols that no laser can realize. This work instead uses the physical laser variables $\Gamma(t)$ and $\nu(t)$ and enforces $0 \leq \Gamma(t) \leq \Gamma_{\max}$. A smoothness penalty and a posteriori slew-rate tests distinguish numerically optimal pulses from controls that can plausibly be produced in the laboratory.

The intended output is a cap-response curve: optimized yield as a function of $\Gamma_{\max}/\gamma$, spanning the tight-cap and saturation regimes. The physical representative is fixed to bosonic ^{88}Sr , using the 1S_0 - 3P_1 intercombination line near 689 nm and an excited Sr_2 photoassociation level. This choice supplies measured linewidth, optical-length,

cloud, and laser scales for translating the dimensionless calculation [8, 9].

1.2. Objectives and contributions

This report is organized around the following deliverables:

1. verify the optical two-channel mapping and its limiting cases;
2. validate a discrete solver for the half-order Volterra dynamics;
3. optimize bounded physical controls with a reproducible multi-start procedure;
4. calibrate the learning-rate schedule and control smoothness, then test convergence, grid dependence, and seed sensitivity;
5. identify cap-binding and cap-insensitive regimes in the molecular yield; and
6. translate one optimized dimensionless protocol into experimentally meaningful intensity, detuning, and duration.

[TO COMPLETE] End the introduction with a short paragraph stating what was actually achieved, including one headline number and the main physical interpretation. Do not repeat the abstract word-for-word.

2. PHYSICAL BACKGROUND

2.1. Two-channel picture

A Feshbach resonance effectively splits the two-body state into an open scattering channel and a closed molecular channel,

$$|\Psi\rangle = |\psi_o\rangle|o\rangle + |\psi_c\rangle|c\rangle. \quad (1)$$

When a laser couples these channels with strength W , the states mix. Near an isolated closed-channel energy level E_c , write $|\psi_c\rangle \simeq b|\phi_c\rangle$. Projecting the stationary Schrödinger equation onto this level gives the closed-channel amplitude

$$b = \frac{\langle\phi_c|W^\dagger|\psi_o\rangle}{E - E_c + i\hbar\gamma/2}, \quad (2)$$

where γ is the decay rate of the excited molecule. The closed-channel admixture, which scales as $P_c \propto |b|^2$, is therefore enhanced near resonance but suppressed by large energy detuning or weak coupling. This mechanism provides the microscopic origin of the resonant, lossy control used below [7].

2.2. Low-energy scattering

The open-channel dynamics at low energy are governed by the s -wave scattering length a_s . Outside the interaction range, the zero-energy scattering wavefunction takes the asymptotic form

$$\psi_0(r) \propto 1 - \frac{a_s}{r}. \quad (3)$$

Purely elastic scattering yields a real scattering length. The molecular decay γ introduced in the two-channel picture instead removes flux irreversibly, requiring a complex scattering length. With the signed-background convention used by the optical map below,

$$a_s = \alpha + i s \beta, \quad \beta \geq 0, \quad s = \text{sgn}(a_{\text{bg}}). \quad (4)$$

The real part α controls the elastic phase shift, while the sign-weighted imaginary part captures irreversible loss into molecular or radiative channels [1]. We employ a zero-range model that enforces this boundary condition while neglecting finite-range microscopic structure inside the interaction region.

[TO COMPLETE] State the regime in which the zero-range and short-time approximations are valid for the chosen gas. Define all density, range, and many-body timescale inequalities that will later be checked.

2.3. Short-time pair amplitude

Consider two identical atoms of mass m in their relative coordinate $\mathbf{r}$. At times short compared with the many-body timescale, their s -wave dynamics are governed by the zero-range Schrödinger equation with the Fermi–Huang pseudopotential:

$$i\hbar \frac{\partial \psi}{\partial t} = \left[-\frac{\hbar^2}{m} \nabla^2 + \frac{4\pi\hbar^2}{m} a_s(t) \delta^{(3)}(\mathbf{r}) \frac{\partial}{\partial r} r \right] \psi. \quad (5)$$

We assume an initially uniform, noninteracting state,

$$\psi(\mathbf{r}, 0) = \frac{1}{\sqrt{\mathcal{V}}}, \quad a_s(0) = 0, \quad (6)$$

where $\mathcal{V}$ is the normalization volume. Thus the initial position-space probability density is uniform, $|\psi(\mathbf{r}, 0)|^2 = 1/\mathcal{V}$.

To solve the short-time dynamics, absorb the time-dependent contact boundary condition into the length-valued short-range amplitude

$$\eta(t) = -4\pi a_s(t) \sqrt{\mathcal{V}} \lim_{r \rightarrow 0} \frac{\partial}{\partial r} [r \psi(\mathbf{r}, t)]. \quad (7)$$

This definition turns the interaction into a time-dependent point source at the origin. Applying Duhamel's formula with the free-particle Green function separates the solution into a $1/r$ spatial singularity and a regular, history-dependent remainder. This finite history is governed by the half-order operator [10]

$$\mathcal{L}_{1/2} \eta(t) = c \lim_{\epsilon \rightarrow 0^+} \left[\int_0^{t-\epsilon} \frac{\eta(\tau)}{(t-\tau)^{3/2}} d\tau - \frac{2\eta(t)}{\sqrt{\epsilon}} \right], \quad (8)$$

where

$$c = \sqrt{\frac{m}{\hbar}} \frac{1}{8\pi^{3/2}\sqrt{i}}. \quad (9)$$

Evaluating the spatial derivative of the Green-function solution near the origin recovers the boundary condition and closes the dynamics as a single Volterra integral equation for $\eta(t)$:

$$\left[\mathcal{L}_{1/2} + \frac{1}{4\pi a_s(t)} \right] \eta(t) = -1. \quad (10)$$

Once $\eta(t)$ is determined, it fixes the short-distance form of the wavefunction through the Bethe–Peierls boundary condition

$$\psi(\mathbf{r}, t) = \frac{\eta(t)}{4\pi\sqrt{\mathcal{V}}} \left[\frac{1}{r} - \frac{1}{a_s(t)} \right] + \mathcal{O}(r). \quad (11)$$

Because the system starts noninteracting, $\eta(0) = 0$, and its subsequent evolution encodes the full history of the controlled scattering length.

2.4. Tan's contact and universal short-time dynamics

Tan's contact $\mathcal{C}$ quantifies short-range pair correlations and appears, for example, in the large-momentum tail

$$n_\sigma(k) \xrightarrow{k \rightarrow \infty} \frac{\mathcal{C}}{k^4}. \quad (12)$$

It connects microscopic pair structure to thermodynamic and dynamical observables [4, 6]. For a gas initially in a noninteracting state, the universal short-time reduction used here writes

$$\mathcal{C}(t) = g_2(0) |\eta(t)|^2, \quad (13)$$

where $g_2(0)$ is the initial local pair density and $\eta(t)$ is determined by two-body short-distance physics [10].

A real square-root ramp, $a_s(t) \propto \sqrt{\hbar t/m}$, gives the scale-invariant onset studied in Ref. [10]. It provides both a conceptual benchmark and a numerical limiting case for the present extension to complex a_s .

2.5. Laser control of the scattering length

Up to this point, $a_s(t)$ is an arbitrary prescribed interaction history. For the optical-control problem, Eq. (10) is analytically continued to a complex scattering length and that history is restricted to values accessible through the laser intensity and detuning. In the adopted two-channel convention, the non-negative control $\Gamma(I)$ is the magnitude of the stimulated width and its physical sign follows $s = \text{sgn}(a_{\text{bg}})$:

$$a_c(I, \nu) = |a_{\text{bg}}| \left[s + \frac{s\Gamma(I)}{-s\Gamma(I) - \nu - i\gamma/2} \right]. \quad (14)$$

Here a_{bg} is the background scattering length, $\Gamma(I) \propto |W|^2 \propto I$ is the stimulated-width magnitude generated by the channel coupling, ν is the laser detuning, and γ is the spontaneous decay rate. Crucially, $\text{Re } a_c$ and $\text{Im } a_c$ are linked by the same two laser controls and cannot be optimized independently [1].

Narrow intercombination transitions reduce the natural linewidth and can therefore produce appreciable scattering changes at moderate intensity. The early ytterbium demonstration changed the zero-energy scattering phase using this strategy [5]. More directly relevant here, optical Feshbach resonances have been measured in a thermal ^{88}Sr gas and used to control the collapse and expansion of an ^{88}Sr condensate [8, 9]. The small negative background scattering length makes even changes of a few Bohr radii dynamically important.

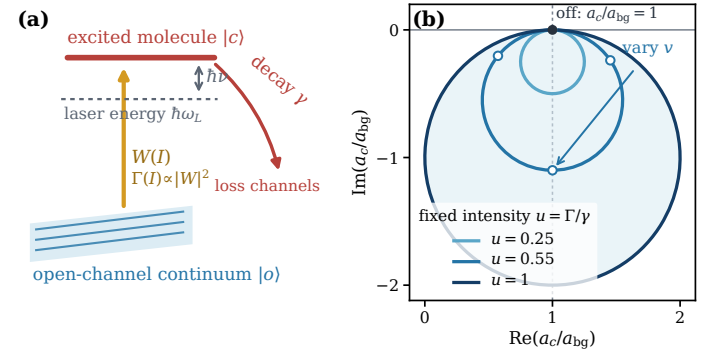


Figure 1. Physical control scheme for the optical Feshbach resonance. (a) A laser couples the open-channel atom-pair continuum to a detuned excited molecular state with matrix element $W(I)$, producing the stimulated width $\Gamma(I) \propto |W|^2$; spontaneous decay at rate γ opens the loss channel. (b) Under Eq. (18), changing detuning ν moves a_c/ℓ_* around a fixed-intensity locus, while changing $u = \Gamma/\gamma$ selects the locus. The elastic and inelastic quadratures are therefore linked rather than independently controllable.

3. MODEL AND CONTROL PROBLEM

3.1. Dimensionless variables

Choose a fixed positive reference time t_* , independent of the as-yet unknown background scattering length. Its associated length is

$$t_* = \frac{m\ell_*^2}{\hbar}, \quad \ell_* = \sqrt{\frac{\hbar t_*}{m}}, \quad \tau = \frac{t}{t_*}, \quad \tilde{a} = \frac{a}{\ell_*}. \quad (15)$$

Optical frequencies are separately measured in units of γ :

$$u(\tau) = \frac{\Gamma(t)}{\gamma}, \quad v(\tau) = \frac{\nu(t)}{\gamma}. \quad (16)$$

The background ratio and its derived sign are

$$r_{\text{bg}} = \frac{a_{\text{bg}}}{\ell_*} = \sqrt{\frac{m}{\hbar t_*}} a_{\text{bg}}, \quad s = \text{sgn}(r_{\text{bg}}). \quad (17)$$

Equation (14) becomes the mapping implemented in the code,

$$\tilde{a}_c(t) = |r_{\text{bg}}| \left[s + \frac{su(t)}{-su(t) - v(t) - i/2} \right]. \quad (18)$$

The absolute-value prefactor is necessary because r_{bg} is stored as a signed ratio: at $u = 0$, Eq. (18) reduces to $\tilde{a}_c = r_{\text{bg}}$. The value $r_{\text{bg}} = 0$ is excluded because its sign is undefined. Its nonzero magnitude and sign are both physical inputs and can be swept without a separate sign parameter. The admissible controls obey

$$0 \leq u(t) \leq u_{\text{max}}, \quad |v(t)| \leq v_{\text{max}}, \quad (19)$$

with $u_{\text{max}} = \Gamma_{\text{max}}/\gamma$.

Once a fixed t_* is selected, each candidate mass and background length maps to a definite r_{bg} through Eq. (17). This retains the physical dependence on m and a_{bg} rather than absorbing it into a species-dependent ruler.

With $t = t_*\tau$, $\eta = \ell_*\tilde{\eta}$, and $t_* = m\ell_*^2/\hbar$, the dimensional factor in Eq. (8) cancels. The dimensionless history-kernel coefficient required by the short-time derivation is therefore

$$\kappa_{\text{th}} = \frac{1}{8\pi^{3/2}\sqrt{i}}. \quad (20)$$

[TO COMPLETE] Derive the contact and yield units, propagate the relatively large uncertainty in a_{bg} , and check that every plotted quantity is labelled either dimensionlessly or in physical units. Keep the relative-coordinate mass convention in Eq. (5) consistent throughout.

[TO COMPLETE] The exact noninteracting start has $\eta(0) = 0$. The current recurrence initializes $\eta_0 = -4\pi a_c(0)$ and uses a separately coded kernel constant. Reconcile that initialization and verify the coded constant against Eq. (20) before reporting production results.

3.2. Molecular-yield objective

Under the sign convention in Eq. (18), the loss-contact relation gives a non-negative product flux proportional to $-s \text{Im}(1/a_c)|\eta|^2$. In the implemented dimensionless normalization, the accumulated objective is

$$\mathcal{J}_{\text{m}}[u, v] = \frac{1}{2\pi} \int_0^T -s \text{Im}\left(\frac{1}{a_c(t)}\right) |\eta(t)|^2 dt. \quad (21)$$

This is the dimensionless, density-like quantity maximized in the current inelastic simulations. It is non-negative for either sign of r_{bg} .

Counting convention to resolve

Equation (21) is a physical yield proxy, but the derivation must establish whether its prefactor counts depleted atoms, lost pairs, or product molecules. A positive global factor does not change the optimizing protocol, but it does change every quoted density, fraction, and molecule number. Until that factor and $g_2(0)$ are restored, report $\mathcal{J}_{\text{m}}$ as a dimensionless objective rather than an absolute molecule count.

3.3. Operational regularization

Piecewise controls can exploit the discretization by developing unrealistically sharp features. To compare physical setups without changing the effective regularization, define $\bar{u} = u/u_{\text{max}}$, $\bar{v} = v/v_{\text{max}}$, and $\theta = t/T$. The code therefore subtracts the normalized finite-difference penalty

$$\mathcal{R}_{\text{sm}} = \frac{1}{\Delta\theta} \sum_{k=0}^{N-1} [\lambda_u(\bar{u}_{k+1} - \bar{u}_k)^2 + \lambda_v(\bar{v}_{k+1} - \bar{v}_k)^2], \quad \Delta\theta = \frac{\Delta t}{T} = \frac{1}{N}. \quad (22)$$

Identical relative pulse shapes therefore receive identical penalties even when T , u_{max} , or v_{max} differ. The optimizer maximizes the regu-

larized score

$$\mathcal{S} = \mathcal{J}_{\text{m}} - \mathcal{R}_{\text{sm}}, \quad (23)$$

while all quoted molecular yields must report the unpenalized $\mathcal{J}_{\text{m}}$ as well. A posteriori slew tests are applied separately to avoid interpreting the regularizer as a hardware model.

4. NUMERICAL METHODS

4.1. Time discretization and L1 history solver

The interval $[0, T]$ is divided into N equal steps, $t_k = k\Delta t$, with controls represented at $N+1$ grid points. The L1 discretization uses the history weights

$$w_{k,j} = \sqrt{k-j} - \sqrt{k-j-1}, \quad 0 \leq j \leq k-2. \quad (24)$$

Defining $q = 2\kappa_{\text{code}}/\sqrt{\Delta t}$ for the code-level kernel prefactor and

$$H_k = \sum_{j=0}^{k-2} w_{k,j}(\eta_{j+1} - \eta_j), \quad (25)$$

the implemented recurrence has the explicit form

$$\eta_k = \frac{-1 + q(\eta_{k-1} - H_k)}{1/[4\pi a_c(t_k)] + q}. \quad (26)$$

Molecular yield is then evaluated with the composite trapezoidal rule. Consistency with the continuous derivation requires the discretization, sign convention, and nondimensionalization to recover $\kappa_{\text{code}} = \kappa_{\text{th}}$ from Eq. (20).

[TO COMPLETE] Complete a constant- or power-law benchmark that verifies the kernel coefficient and initialization independently of optimization. Then give the formal convergence order for the solution regularity, computational complexity, and measured convergence from Section 5.

4.2. Bounded physical controls

Unconstrained optimizer variables $x_k, y_k \in \mathbb{R}$ are mapped smoothly into the admissible laser domain:

$$u_k = u_{\text{max}} \text{sigmoid}(x_k), \quad v_k = v_{\text{max}} \tanh(y_k). \quad (27)$$

This parameterization makes every iterate physical with respect to the amplitude bounds and remains differentiable for automatic differentiation. Saturation of either map can, however, reduce gradients; cap-usage diagnostics are therefore retained.

4.3. Gradient optimization

The calculation follows the logic of GRAPE: discretize the controls, differentiate the terminal performance with respect to every time slice, and update the complete pulse [3]. JAX differentiates through the scattering map, Volterra scan, quadrature, and regularizer. Batched Adam updates are compiled and vectorized, with 64-bit arithmetic used for production calculations.

The learning rate is changed in stages without resetting Adam's moment estimates. Stability is evaluated at fixed checkpoint intervals independently of those stages. For run r , the reported solution is the best finite iterate according to $\mathcal{S}$ when smoothing is active and according to $\mathcal{J}_{\text{m}}$ otherwise.

4.4. Multi-start strategy and convergence tests

The current workflow combines 16 structured u - v initializations with 24 reproducible random Fourier initializations. Selected optimized controls are perturbed with smooth, zero-mean Fourier noise to probe local robustness. Seed sensitivity is treated as a population diagnostic, not as a stopping condition for an individual run.

At present, the internal stopping test measures motion in the raw optimizer coordinates $z \in \{x, y\}$ over a checkpoint block of M steps,

$$\epsilon_z^{\text{raw}}(k) = \frac{\|z_k - z_{k-M}\|_2}{\|z_k\|_2 + \epsilon}. \quad (28)$$

Because the sigmoid and tanh maps distort this distance near their bounds, final reporting should additionally use bounded, range-normalized changes

$$\epsilon_u = \frac{\|u_k - u_{k-M}\|_2}{\sqrt{N+1} u_{\max}}, \quad \epsilon_v = \frac{\|v_k - v_{k-M}\|_2}{2\sqrt{N+1} v_{\max}}. \quad (29)$$

A run is accepted only after the regularized score and every active control pass their tolerances for the required number of consecutive checkpoints. The score test, raw-coordinate stopping test, and bounded-control diagnostic must be labelled separately in figures and archived data.

Table 1. Baseline $N = 100$ numerical protocol encoded in the current calibration workflow. Confirm or replace these entries when production settings are frozen.

Quantity	Current value
Grid intervals N	100
Initial learning rate	10^{-2}
Stage schedule	(5000, 1), (5000, 0.5), (7500, 0.5), (7500, 0.5)
Stability interval	500 steps
Consecutive passes	3
Score tolerance	10^{-5}
Raw-control tolerances	10^{-3} for x and y
Initializations	16 structured +24 Fourier
Smoothness scan	17 log-spaced points over 10^{-4} –1
Shared slew parameter	0.05
Arithmetic	JAX float64

Learning-rate multipliers compound between stages. The smoothness selection currently weights valid-run yield by 0.7 and roughness by 0.3.

4.5. Calibration sequence and reproducibility

The production methodology fixes hyperparameters in a prescribed order:

1. select the learning-rate schedule at $N = 100$;
2. calibrate the shared smoothness against yield and slew validity;
3. sweep the initial intensity cap using those frozen settings; and
4. perform the final $u_{\max}$ sweep and robustness checks.

Numerical data, metadata, and figures are stored separately so that plots can be recreated without rerunning the optimizer.

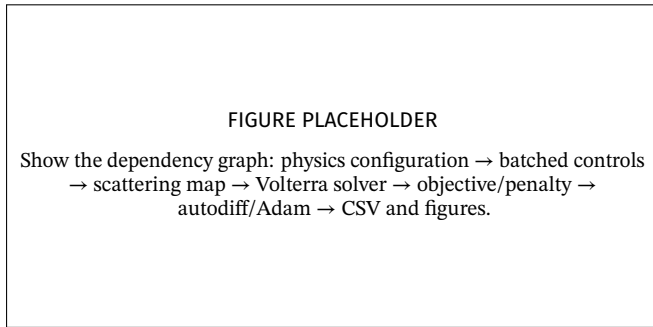


Figure 2. Computational workflow and artifact flow. Annotate the source-labelled directories used for configurations, results, figures, and logs.

5. VERIFICATION AND VALIDATION

5.1. Analytical checks of the scattering map

Equation (18) should be checked in three limits before optimization: $u \rightarrow 0$ gives $a_c \rightarrow a_{bg}$; large $|v|$ suppresses the optical shift and loss; and the near-resonant response reproduces the expected dispersive real part and absorptive imaginary part. These checks also establish the sign conventions for v and $\text{Im } a_c$.

Table 2. Minimum validation matrix. Replace the final column with measured errors and pass criteria.

Test	Expected behaviour	Result
No laser	$a_c/a_{bg} \rightarrow 1$	[insert]
Far detuned	optical term $\rightarrow 0$	[insert]
Complex sign	$\text{Im } a_c < 0$	[insert]
Constant a_s	known/reference $\eta(t)$	[insert]
Square-root ramp	universal onset law	[insert]
Grid refinement	convergent $\mathcal{J}_m$ and pulse	[insert]
Autodiff gradient	finite-difference agreement	[insert]

5.2. Volterra solver and quadrature convergence

Optimization stability at fixed N is not evidence of time-grid convergence. For a sequence such as $N = 100, 200, 400, 800, 1000$, define the relative yield difference

$$\epsilon_N = \frac{|\mathcal{J}_m^{(N)} - \mathcal{J}_m^{(N_{\text{fine}})}|}{\max(|\mathcal{J}_m^{(N_{\text{fine}})}|, 10^{-12})}. \quad (30)$$

If no analytic reference is available, the difference between the two finest grids is an error estimate, not a rigorous bound. Report the error in $\eta(t)$ where a reference is available and whether qualitative control features persist. Because optimization can exploit grid-specific structure, use both routes: evaluate one fixed interpolated pulse on every grid, and independently reoptimize to comparable convergence at every N .

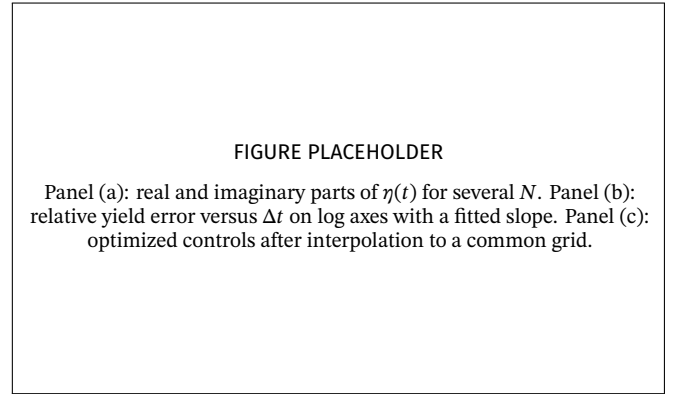


Figure 3. Validation and grid-convergence tests for the L1 Volterra solver and optimized objective. State the reference solution and error norm in the caption.

5.3. Gradient and baseline checks

On a reduced grid, compare automatic derivatives of $\mathcal{S}$ with centered finite differences for randomly selected control components. Then reproduce the real-scattering-length or unconstrained baseline associated with the universal onset problem before introducing the physical OFR mapping.

[TO COMPLETE] Give numerical tolerances, hardware, software versions, and the commit identifier. A statement such as “the curves look similar” is not a validation criterion; quote norms and relative errors.

6. RESULTS

6.1. Learning-rate and convergence calibration

Present the evidence used to select the staged Adam schedule. Separate the unpenalized physical objective $\mathcal{J}_m$, smoothness penalty $\mathcal{R}_{\text{sm}}$, and regularized score $\mathcal{S}$, and show the checkpoint-based convergence criteria for the best and median initializations.

Provisional optimizer observation

FIGURE PLACEHOLDER

Recommended panels: $\mathcal{J}_m$, $\mathcal{R}_{sm}$, and $\mathcal{S}$ histories with learning-rate boundaries; score stability; bounded ϵ_u and ϵ_v from Eq. (29); final-yield distribution including failures. Overlay all converged controls faintly, with the median or best protocol emphasized.

Figure 4. Learning-rate calibration and convergence diagnostics at $N = 100$. Vertical markers should identify stage changes and horizontal markers the stopping tolerances. Report $n_{\text{success}}/40$, define failure before inspecting outcomes, and display rather than silently remove any zero-yield run.

The project log records an exploratory schedule with one (5000, 1) stage followed by nine (5000, 0.5) stages. From an initial rate 10^{-2} , the nine cumulative halvings give a final rate 1.95×10^{-5} without resetting Adam's moments. Under the metric then in use, the objective change reached approximately 10^{-10} while the two control changes remained just above 10^{-3} . Most starts approached similar controls, but one reached zero yield; milder annealing retained a wider family of controls. These observations must be reproduced from archived run data before being promoted to quantitative results. They suggest a shallow or poorly conditioned optimum, not an exactly flat objective.

Use the fully annealed ensemble for the primary yield estimate once reproduced; use milder schedules only to diagnose the landscape. Numerical precision in the reported yield is limited by the grid-convergence study, even when optimizer stability is much smaller.

[TO COMPLETE] Archive and reproduce the ten-stage observation, then state which schedule was selected and why. Diagnose any zero-yield start by checking $u \simeq 0$, saturation of raw variables, gradient norms, and non-finite values; unless it is shown to be a valid stationary solution, classify it as an optimization failure.

6.2. Shallow-landscape diagnosis

Similar values of $\mathcal{J}_m$ need not imply identical controls. Let (u^*, v^*) be the best converged protocol and define, for run i ,

$$d_u^{(i)} = \frac{\|u^{(i)} - u^*\|_2}{\sqrt{N+1} u_{\max}}, \quad d_v^{(i)} = \frac{\|v^{(i)} - v^*\|_2}{2\sqrt{N+1} v_{\max}},$$

$$d_{uv}^{(i)} = \sqrt{(d_u^{(i)})^2 + (d_v^{(i)})^2}, \quad \Delta \mathcal{J}^{(i)} = \frac{\mathcal{J}_m^* - \mathcal{J}_m^{(i)}}{|\mathcal{J}_m^*|}. \quad (31)$$

A broad range of d_{uv} at small $\Delta \mathcal{J}$ supports a shelf of nearly equivalent solutions. Plot both $\mathcal{J}_m$ and $\mathcal{S}$: flat $\mathcal{J}_m$ but curved $\mathcal{S}$ means the regularizer selects a smooth representative, whereas flatness in both gives stronger evidence of a shallow physical landscape.

Confirm this with no-reoptimization line scans in raw optimizer space, $z(\lambda) = (1-\lambda)z^* + \lambda z^{(i)}$. Compare not only the controls but the induced $a_c(t)$, $|\eta(t)|^2$, and flux $\text{Im}(1/a_c)|\eta|^2$: visibly different laser waveforms may be physically equivalent after the nonlinear scattering map and history dynamics.

6.3. Smoothness and slew calibration

The shared coefficient $\lambda_u = \lambda_v = \lambda$ is scanned logarithmically. The choice must balance physical yield against roughness while satisfying separate normalized slew limits for both controls. Report all run-level points, not only medians, so that failure fractions and outliers remain visible.

[TO COMPLETE] Report the selected λ , valid-run fraction, loss in unpenalized yield relative to weak regularization, and sensitivity of the choice to the 0.7/0.3 selection weighting.

6.4. Optimized laser-constrained protocol

For one representative cap, show how the optimized physical controls generate the complex scattering length, pair amplitude, instantaneous molecular flux, and cumulative yield. This is the key figure for explaining the mechanism rather than only the performance.

[TO COMPLETE] Explain the pulse chronologically. Distinguish observations visible in the data from physical interpretations such as contact preparation, conversion, or quantum-Zeno-like suppression.

6.5. Yield versus available laser power

The master result is obtained by sweeping $u_{\max} = \Gamma_{\max}/\gamma$ over approximately three decades while holding the validated numerical methodology fixed. Both best and median multi-start yields should be shown. The normalized cap usage

$$r_u = \max_k \frac{u_k}{u_{\max}} \quad (32)$$

distinguishes power-limited solutions from those that leave additional intensity unused.

Table 3. Quantitative comparison of the regimes in Fig. 7.

Regime	$u_{\max}$ range	Observed behaviour
Tight cap	[insert]	[yield scaling and r_u]
Crossover	[insert]	[pulse/mechanism change]
Saturation	[insert]	[plateau criterion]

[TO COMPLETE] Quote the fitted low-cap scaling if supported, the crossover cap, the plateau yield, and the incremental gain from the largest tested cap. Do not infer a true asymptote from too narrow a sweep.

6.6. Robustness and uncertainty

Robustness should cover at least initialization seed, grid resolution, convergence tolerance, smoothness choice, and modest perturbations to the final controls. Treat zero or non-finite yield as a reported failure category rather than filtering it from the seed statistics. Where experimental parameters are uncertain, repeat the final protocol for representative variations in γ , intensity calibration, and detuning offset.

FIGURE PLACEHOLDER

Combine the generated calibration figures: (a) normalized u slew versus λ ; (b) normalized v slew versus λ ; (c) valid-run median yield and roughness or weighted selection score. Mark the selected λ .

Figure 5. Smoothness calibration under the configured slew constraints. Define the slew metric, valid-run criterion, and selection rule in the caption.

FIGURE PLACEHOLDER

Recommended panels: (a) $u(t)$ with $u_{\max}$; (b) $v(t)$ with detuning bounds; (c) trajectory of a_c/a_{bg} in the complex plane; (d) $|\eta(t)|^2$; (e) instantaneous $\text{Im}(1/a_c)|\eta|^2/(2\pi)$; (f) cumulative $\mathcal{J}_m(t)$.

Figure 6. Representative optimized protocol and its physical response. Use a common normalized time axis, mark cap-active intervals, and identify which historical features build contact before molecular conversion.

FIGURE PLACEHOLDER

Summarize seed spread and robustness: final-yield distribution with failures; ΔI versus d_{uv} ; raw-space line scans of $\mathcal{J}_m$ and $\mathcal{S}$; yield after smooth Fourier perturbations; and relative changes under grid or parameter variations.

the optical length ℓ_{opt} rather than a universal constant ξ :

$$\ell_{\text{opt}} = \beta I, \quad \Gamma_{\text{stim}} = 2k_r \ell_{\text{opt}} \Gamma_{\text{mol}}, \quad \beta = 2.2 \times 10^4 \frac{a_0}{\text{W cm}^{-2}}, \quad (33)$$

where k_r is the relative collision wave number and $\Gamma_{\text{mol}} = 2\pi \times 15 \text{ kHz}$ [9]. If the model quantities are identified as $\Gamma = \Gamma_{\text{stim}}$, $\gamma = \Gamma_{\text{mol}}$, and $\nu = \Delta$, then

$$I_{\max} = \frac{u_{\max}}{2k_r \beta}, \quad \frac{\Delta(t)}{2\pi} = (15 \text{ kHz}) v(t). \quad (34)$$

Thus an intensity cap cannot be assigned until k_r is fixed consistently from the simulated gas. By contrast, the current $v_{\max} = 50$ maps directly to $|\Delta|/2\pi = 0.75 \text{ MHz}$, within the experimentally explored range.

7.2. Experimentalist's recipe

Use $I = 0.057 \text{ W cm}^{-2}$, waist $725 \mu\text{m}$, and detunings of order 1 MHz as a literature-anchored comparison point, not yet as the optimized recommendation. The quoted beam corresponds to approximately 0.47 mW if I is the peak intensity of a Gaussian beam. Once the fixed reference time t_* has been selected, the reported 1.2- and 4-ms exposures must be converted using $\tau = t/t_*$; they cannot be represented by silently setting the solver's $T = 1$ equal to one millisecond.

The experimental isolated-resonance model contains the fitted loss enhancement $\eta_{\text{loss}} = 19.5$ and a detuning-dependent background loss not present in Eq. (14) [9]. The subscript distinguishes this empirical loss parameter from the pair amplitude $\eta(t)$ in Eq. (13). The ^{88}Sr prescription is therefore conditional on first verifying the mapping between the measured model and the implemented $a_c(u, v)$, especially the roles of Γ_{stim} , Γ_{mol} , and the phenomenological loss terms.

Figure 8. Numerical, landscape, and control robustness of the optimized yield. Give the total and successful run counts, define every perturbation distribution, and distinguish optimizer spread from discretization error.

7. EXPERIMENTAL TRANSLATION

7.1. Fixed representative: bosonic strontium-88

The physical reference is an ^{88}Sr condensate in the $^1S_0 + ^1S_0$ entrance channel, optically coupled near the $^1S_0 + ^3P_1$ molecular asymptote. The atomic intercombination line is near 689 nm ; its measured ^{88}Sr frequency is $434\,829\,121\,311(10) \text{ kHz}$ [2]. The condensate experiment used the second-least-bound excited molecular level, with binding energy $h \times 24 \text{ MHz}$ below this asymptote [9].

The measured intensity calibration is naturally expressed through

FIGURE PLACEHOLDER

Panel (a): optimized $\mathcal{J}_m$ versus $u_{\max}$ on a logarithmic cap axis, with best, median, and run spread. Add the unconstrained/universal benchmark where justified. Panel (b): median and maximum r_u . Shade tight-cap, crossover, and saturation regimes.

Figure 7. Optimized molecular yield as a function of the available optical coupling. Define the statistical summary, cap-binding threshold, uncertainty band, and criterion used to identify saturation.

Table 4. Fixed ^{88}Sr reference parameters. “Derived” values use the central reported quantities and are not additional measurements.

Parameter	Value	Source
Isotope and channel	Bosonic ^{88}Sr ; $^1S_0 + ^1S_0 \rightarrow ^1S_0 + ^3P_1$	[8, 9]
Atomic line	689 nm; 434 829 121 311(10) kHz	[2]
Background length	$a_{\text{bg}} = -1.4(6)a_0$; the condensate paper uses the rounded value $-2a_0$	[8, 9]
Molecular reference	Second-least-bound level at -24 MHz; $\Gamma_{\text{mol}}/2\pi = 15$ kHz	[9]
Optical coupling	$\ell_{\text{opt}}/I = 2.2 \times 10^4 a_0/(\text{W cm}^{-2})$; at 0.057 W cm^{-2} , $\ell_{\text{opt}} = 1.25 \times 10^3 a_0$	[9]; derived
Demonstrated controls	$ \Delta /2\pi$ up to 10 MHz, or $ \Delta /\Gamma_{\text{mol}} \simeq 667$; typical $a = 20a_0$ at +1 MHz; changes up to $a_{\text{opt}}/a_{\text{bg}} = \pm 10$	[9]
Condensate	$N_0 \simeq 7000$, peak density $n_0 = 1 \times 10^{15} \text{ cm}^{-3}$, initial size $\sigma_0 = 0.8 \mu\text{m}$, trap mean frequency $2\pi \times (60 \pm 5) \text{ Hz}$	[9]
Laser and timing	Waist $725 \mu\text{m}$; example intensity 0.057 W cm^{-2} ; 1.2- and 4-ms exposures. The intensity corresponds to 0.47 mW for an on-axis Gaussian convention	[9]; derived
Loss scale	$K_{\text{in}} \sim 10^{-12} \text{ cm}^3 \text{ s}^{-1}$, millisecond sample lifetimes, fitted enhancement $\eta_{\text{loss}} = 19.5$	[9]

[TO COMPLETE] Calculate k_r for the final cloud model, convert every optimized $u(t)$ through Eq. (34), and compare the resulting peak power, modulation bandwidth, and duration with Table 4. Re-run the dynamics with the measured enhanced and background loss before claiming a quantitative ^{88}Sr molecule yield.

8. DISCUSSION

8.1. Physical interpretation

[TO COMPLETE] Explain why the optimal history outperforms simple constant, linear, square-root, or bang-bang trial protocols. Relate the protocol to the competition between building contact and converting correlated pairs into products. If invoking a quantum-Zeno mechanism, show the diagnostic that supports it.

Compare the cap sweep with the scale-invariant real- a_s onset of Ref. [10]. Make explicit which aspects remain universal and which are introduced by the optical map, finite decay rate, detuning bound, and control regularization.

8.2. Comparison with baseline protocols

Table 5. Performance comparison at a common duration and cap.

Protocol	$\mathcal{J}_m$	Relative to optimum
Best constant controls	[insert]	[insert]
Simple ramp	[insert]	[insert]
Square-root-inspired	[insert]	[insert]
Unconstrained baseline	[insert]	[insert]
Optimized OFR controls	[insert]	1

8.3. Limitations

The final report should assess at least the following limitations:

- validity of the universal short-time, zero-range, and dilute-gas approximations;
- neglect of depletion, many-body feedback, trapping inhomogeneity, and additional loss channels;
- accuracy and convention dependence of the two-channel OFR map;
- numerical bias from finite N , smoothness selection, bounded parameterization, and local optimization;
- distinction between the finite-difference slew diagnostic and a measured laser transfer function;
- collision-momentum dependence in translating Γ to 689-nm intensity for ^{88}Sr ; and
- omission of the measured $\eta_{\text{loss}} = 19.5$ enhancement and far-wing background loss from the implemented optical map.

[TO COMPLETE] For each material limitation, state the likely direction of bias and one concrete test or future extension.

8.4. Mathematical and computational extensions

After the baseline calculation is validated, the Hessian spectrum of $\mathcal{S}$ can quantify poorly constrained control directions more directly than endpoint scatter alone. Hessian-vector products or a reduced-basis eigenspectrum are preferable to forming the full dense Hessian. Quasi-Newton polishing can then be compared fairly with the long Adam anneal. A robust-control extension should maximize an ensemble average, or a risk-sensitive alternative, over calibrated Gaussian laser-amplitude and detuning noise. Physics-informed neural networks

FIGURE PLACEHOLDER

Primary axis: predicted ^{88}Sr molecular yield/fraction versus peak 689-nm intensity, with a power axis for $w = 725\ \mu\text{m}$. Mark $0.057\ \text{W cm}^{-2}$ as the literature comparison and the optimized operating point separately. Inset: intensity and detuning waveforms in ms and MHz relative to the -24-MHz PA level.

Figure 9. Experimental prescription for ^{88}Sr . Propagate uncertainty in a_{bg} , ℓ_{opt}/I , density, and collision momentum; state whether the prediction is a dimensionless objective, density, fraction, or absolute molecule number.

remain a possible surrogate for the nonlocal dynamics, but only after their accuracy and cost are benchmarked against the L1 solver.

9. CONCLUSION

[TO COMPLETE] Write this section only after the figures are final. In one compact paragraph: answer the central question; quote the optimal/plateau yield and crossover cap; state the representative laboratory requirement; summarize robustness; and give the most important limitation. Follow with 2–3 specific next steps.

The completed conclusion should distinguish the scientific result—how a physical laser constraint shapes the optimal molecular yield—from the computational deliverable of a validated, reproducible JAX workflow.

■ AUTHOR CONTRIBUTIONS

[TO COMPLETE] Use the CRediT taxonomy if appropriate. State the student’s work and the supervisor’s roles in conceptualization, methodology, software, analysis, supervision, and writing.

■ ACKNOWLEDGEMENTS

[TO COMPLETE] Acknowledge the supervisor, collaborators, CUHK programme support, computing resources, and any funding.

■ DATA AND CODE AVAILABILITY

The reusable package under `src/optical_feshbach_control/` separates calculations, analysis, plotting, and artifact I/O. Versioned run settings belong in `configs/`; numerical artifacts in `results/`; regenerated graphics in `figures/`; and queue logs in `logs/`. The $N = 100$ calibration notebooks and non-interactive smoothness workflow record the ordered methodology used in this report.

[TO COMPLETE] Add the repository URL or archive DOI, exact commit hash, environment file, commands needed to reproduce every final figure, and any access restrictions.

A. NONDIMENSIONALIZATION AND CONVENTIONS

[TO COMPLETE] Extend the dimensional audit above to the contact and molecular-yield normalizations. Track the sign of the imaginary scattering length and all factors of 2 , π , $\hbar$, and mass, then verify the ledger against the implementation.

Table 6. Symbol and unit ledger for the final derivation.

Symbol	Meaning	Unit/scale
a_{bg}	background scattering length	$-1.4(6)a_0$
a_{c}	complex scattering length	length; scale $ a_{\text{bg}} $
γ	molecular decay rate	$2\pi \times 15\ \text{kHz}$
Γ	stimulated optical width	$2k_r \ell_{\text{opt}} \Gamma_{\text{mol}}$
ν	detuning from -24-MHz PA level	angular frequency
$\eta(t)$	short-range pair amplitude	length; scale $ a_{\text{bg}} $
$\mathcal{C}$	Tan-contact density	inverse length ⁴
$\mathcal{J}_{\text{m}}$	current yield objective	dimensionless; prefactor unresolved

B. DISCRETE SOLVER DERIVATION

[TO COMPLETE] Derive the L1 weights and recurrence in Eqs. (24)–(26). State how the endpoint singularity is handled, why the newest increment enters the denominator, and the memory/time complexity of the scan.

C. ADDITIONAL OPTIMIZATION DIAGNOSTICS

[TO COMPLETE] Place full per-seed convergence plots, zero-yield diagnostics, raw- versus bounded-control stability, alternative schedules, raw-space line scans, smoothness sensitivity, and higher-resolution cap-sweep tables here. Keep only decision-relevant summaries in the main text.

D. REPRODUCTION CHECKLIST

1. Record the repository commit and Python environment.
2. Freeze the learning-rate, smoothness, convergence, and initialization settings.
3. Save run-level CSV/JSON data before plotting.
4. Label every result by grid, seed, cap, and configuration.
5. Regenerate report figures from saved numerical artifacts.
6. Confirm that all reported headline values can be read from an archived table.

E. FIXED REFERENCE TIME AND SWEEPABLE BACKGROUND RATIO

A reference time $t_{\star} > 0$ is fixed across candidate physical setups. It defines a positive length $\ell_{\star} = \sqrt{\hbar t_{\star}/m}$, not a removable physical scattering length. Consequently

$$r_{\text{bg}} = \frac{a_{\text{bg}}}{\ell_{\star}} = \sqrt{\frac{m}{\hbar t_{\star}}} a_{\text{bg}} \quad (35)$$

remains a signed model parameter. Sweeping r_{bg} represents candidate values of a_{bg} and m at the fixed t_* .

With $s = \text{sgn}(r_{bg})$, the code evaluates

$$\tilde{a}_s = |r_{bg}| \left[s + \frac{su}{-su - v - i/2} \right]. \quad (36)$$

The prefactor is $|r_{bg}|$ because r_{bg} itself is signed; this guarantees the far-detuned or zero-width limit $\tilde{a}_s = r_{bg}$. The value $r_{bg} = 0$ is invalid because s would be undefined, and there is no independent sign parameter in the configuration.

Finally, regularization is applied to the normalized curves $u/u_{\max}$ and $v/v_{\max}$ over normalized time t/T . Therefore a common relative pulse shape has one smoothness cost independent of its dimensional duration, linewidth conversion, and absolute control caps.

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